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Reg. No. 1 2 1 0 2

Time : 3.00 hrs.

# Quarterly Examination - 2025

## BIOLOGY

Max. Marks : 70

### PART - I (BIO-BOTANY)

### SECTION - 1

**I. Choose the correct answer.** 8x1=8

1. Which of the following represent mega gametophyte.
   a) Ovule b) Embryo sac c) Nucellus d) Endosperm
2. Extra nuclear inheritance is a consequence of presence of genes in
   a) mitochondria and chloroplasts b) Endoplasmic reticulum and mitochondria c) Ribosomes and chloroplasts d) Lysosomes and ribosomes
3. Genetic Engineering is a) making artificial genes b) hybridization of DNA of one organism to that of the other c) production of alcohol by using micro organisms d) making artificial limbs, diagnostic instruments such as ECG, EEG, etc.
4. An antibiotic gene in a vector usually helps in the selection of
   a) competent cells b) transformed cells c) recombinant cells d) none of the above
5. The prevention of large scale loss of biological integrity
   a) Bio-patent b) Bio-ethics c) Bio-safety d) Bio-fuel
6. Virus free plants are developed from
   a) organ culture b) meristem culture c) protoplast culture d) cell suspension culture
7. The term crossing over was coined by a) Montgomery b) T.Boveri c) W.S.Sutton d) T.H.Morgan
8. E.Coli is the mostly used organism for gene cloning, because a) it is easy to handle
   b) it is growing easily under optional condition c) it is the safe organism d) all the above

### SECTION - 2

**II. Answer any four of the following questions.** 4x2=8

9. Write short note on Pollen Kitt.
10. Distinguish mound layering and air layering.
11. What is back cross?
12. Give any two physical mutagens.
13. Name the chemicals used in gene transfer.
14. Define: 'Totipotency'.

### SECTION - 3

**III. Answer any three of the following questions. Q.No.19 is compulsory.** 3x3=9

15. Draw and label the parts of T.S. of mature anther.
16. What are the reasons for Mendel's successes in his breeding experiment?
17. Write any three uses of gene mapping.
18. What do you know about the word 'pBR 322'?
19. Given an account on Cryopreservation.

### SECTION - 4

**IV. Answer all the questions.** 2x5=10

20. With a suitable diagram explain the structure of an ovule. **(OR)**
    Explain incomplete dominance with suitable example.
21. Mention the application of Bio-technology. **(OR)**
    Explain the steps involved in protoplast culture.



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## PART - II (BIO-ZOOLOGY)

### SECTION - A

Note: (i) Answer all the questions (ii) Choose the most appropriate answer from the given four alternatives and write the option code and corresponding answer. 8x1=8

1. Match the following type of binary fission and the examples correctly.

| Types of binary fission | Example |
|---|---|
| I) Repeated binary fission | A) Ceratium |
| II) Oblique binary fission | B) Planaria |
| III) Longitudinal binary fission | C) Vorticella |
| IV) Transverse binary fission | D) Euglena |

| | I | II | III | IV | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| a) | A | D | B | C | c) | B | C | A | D |
| b) | C | A | D | B | d) | D | B | C | A |

2. The mature sperms are stored in the _____.
   a) Seminiferous tubules b) Vas deferens c) Seminal Vesicle d) Epididymis
3. A contraceptive pill prevents ovulation by a) blocking fallopian tube
   b) stimulating release of FSH and LH c) Inhibiting release of FSH and LH
   d) Causing immediate degeneration of released ovum.
4. Co-dominant blood group is _____ a) AB b) A c) B d) O
5. Which of the following is the correct sequence of event with reference to the Central dogma?
   a) Replication, Transcription, Translation b) Transcription, Translation, Replication
   c) Transcription, Replication, Translation d) Duplication, Translation, Transcription
6. Which period was called "Age of Fishes"? a) Permian b) Triassic c) Devonian d) Ordovician
7. AIDS virus has _____
   a) Single stranded RNA b) Double stranded RNA c) Single stranded DNA d) Double stranded DNA
8. The most common substrate used in distilleries for the production of ethanol is _____.
   a) Ground gram b) Corn meal c) Soya meal d) Molasses.

### SECTION - B

**Answer any four of the following.** **4x2=8**

9. What is plasmotomy?
10. What is Menstrual Cycle?
11. Mention the symptoms of Down's Syndrome.
12. Why the human genome project is called a mega project?
13. What are interferons? Mention their role.
14. What is Bioremediation?

### SECTION - C

**Answer any three of the following. Question No.19 is compulsory.** **3x3=9**

15. Draw a labelled sketch of human ovum.
16. Differentiate – Template strand and coding strand.
17. Write the main objections to Darwinism.
18. List the causative agent, mode of transmission and symptoms for Diphtheria and Typhoid.
19. Explain the marriage between normal visioned man and colour blind woman.

### SECTION - D

**Answer all the questions.** **2x5=10**

20. a) What is parthenogenesis? Explain the types of parthenogenesis. **(OR)**
    b) Explain the Barrier methods of Birth Control.
21. a) Explain the formation of a nucleosome. **(OR)**
    b) Explain the structure of Immunoglobulin with suitable diagram.



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